Review


```
"Summary": "The paper investigates the impact of weight initialization
strategies on the grokking phenomenon in Transformer models, focusing on
arithmetic tasks in finite fields. It compares five initialization methods
(PyTorch default, Xavier, He, Orthogonal, and Kaiming Normal) using a small
Transformer architecture. The study reveals significant differences in
convergence speed and generalization capabilities across initialization
strategies, with Xavier and Orthogonal initializations showing superior
performance.",
"Strengths": [
    "Addresses an intriguing and underexplored phenomenon in deep
learning.",
    "Provides a systematic comparison of multiple weight initialization
strategies.",
    "Includes rigorous empirical analysis and statistical validation.",
    "Offers practical guidelines for initialization in similar learning
scenarios."
],
"Weaknesses": [
    "The scope is limited to small Transformer models and arithmetic tasks,
which may not generalize well to larger models or more complex tasks.",
    "The paper lacks deeper theoretical insights into why certain
initialization strategies perform better.",
    "The clarity of the experimental setup and the integration of figures
and tables could be improved.",
    "The implications for broader Transformer applications and potential
societal impacts are not sufficiently addressed."
],
"Originality": 3,
"Quality": 3,
"Clarity": 3,
"Significance": 3,
"Questions": [
    "Can the authors provide more theoretical explanations for why certain
initialization methods perform better?",
    "How do the findings translate to more complex, real-world tasks beyond
simple arithmetic operations?",
    "Can the clarity of the figures and tables be improved, and can key
graphs be better integrated into the text?",
    "What are the potential negative societal impacts of the findings?"
],
"Limitations": [
    "The study is limited to small Transformer models and arithmetic tasks,
which may not fully represent the complexity of real-world problems.",
    "The paper lacks a deeper theoretical understanding of the observed
phenomena.",
    "The potential negative societal impacts of the findings are not
addressed."
],
"Ethical Concerns": false,
"Soundness": 3,
"Presentation": 3,
"Contribution": 3,
"Overall": 5,
```

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"Confidence": 4,
"Decision": "Reject"
```

D.8. Grokking Accelerated: Layer-wise Learning Rates for Transformer Generalization

This idea was proposed in the 22nd iteration of a Sonnet 3.5 run.

Idea

```
"Name": "layerwise_lr_grokking",
"Title": "Layer-wise Learning Rate Grokking: Assessing the impact of layer-
wise learning rates on the grokking phenomenon",
"Experiment": "Modify the `run` function to implement layer-wise learning
rates. Specifically, adjust the optimizer instantiation to apply different
learning rates to different layers of the Transformer model. Define three
groups: 1) Embedding layers with a small learning rate (e.g., 1e-4), 2)
Lower Transformer layers with a moderate learning rate (e.g., 1e-3), 3)
Higher Transformer layers with a larger learning rate (e.g., 1e-2). Use
PyTorch's parameter groups feature to assign these learning rates. Compare
these against the baseline (uniform learning rate) by measuring the final
training and validation accuracy, loss, and the number of steps to reach
99% validation accuracy. Evaluate the results for each dataset and seed
combination.",
"Interestingness": 9,
"Feasibility": 8,
"Novelty": 9,
"novel": true
```

Link to code: https://github.com/SakanaAI/AI-Scientist/tree/main/example_papers/layerwise_lr_grokking.